

Question	Answer	Mark	Mark scheme	Additional guidance
19	31.0	P1	for $\tan 35 = BE \div 15$ or $BE = 10.5(0\dots)$ OR finding the length $DM = \frac{2}{5} \times 15 (= 6)$ or $MA = \frac{3}{5} \times 15 (= 9)$ or 6:9 OR showing the required angle on a diagram eg with an arc	$MB = \sqrt{9^2 + 15^2} = \sqrt{306}$ (=17.4(9...) or 17.5) $BE = 15 \times \tan 35$ (=10.5(0...)) $AE = 15 \div \cos 35$ (=18.3(1...)) $ME = \sqrt{9^2 + 18.31\dots^2} = \sqrt{416. (3 \dots)}$ (=20.4(0...))
		P1	for $MB = \sqrt{15^2 + "9"^2}$ or $\sqrt{306}$ or 17.4(9....) OR $ME = \sqrt{"9"^2 + "18.3(1 \dots)"}^2$ or $\sqrt{416. (3 \dots)}$ or 20.4(0...)	Check diagram for working
		P1	for using appropriate trigonometry ratio to set up an equation in angle EMB eg $\tan \theta = "10.5(0\dots)" \div "17.4(9\dots)"$ or $\cos \theta = "17.4(9\dots)" \div "20.4(0\dots)"$ or $\sin \theta = "10.5(0\dots)" \div "20.4(0\dots)"$	
		A1	for answer in the range 30.9 to 31	If an answer is shown in the range in working and then incorrectly rounded award full marks.